

TILAK RAGHUNANDAN G

AI/ML Engineer | Computer Vision | NLP & LLMs | Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year AI & Machine Learning engineering student (graduating January 2027) with hands-on experience building RAG pipelines, LLM-integrated tools, and computer vision models for medical imaging, developed through an industry internship, national-level hackathons, and independent research projects. Proven track record with 3 podium finishes at state- and national-level hackathons and a 95%-accuracy medical diagnostics model. Strong full-stack foundation (React.js, Node.js, MySQL, MongoDB) complements deep learning research skills. Seeking an AI/ML Engineering, Computer Vision, or NLP role to contribute from day one.

TECHNICAL SKILLS

Programming Languages: Python (Primary), Java (Basic), C, JavaScript

AI, GenAI & LLMs: RAG Pipelines, LLM Integration, Prompt Engineering, AI Agents, Embeddings, Semantic Search, Vector Databases, Hugging Face (Transformers)

Machine Learning & Computer Vision: PyTorch, Scikit-learn, CNNs, Transfer Learning, U-Net, ResNet, VGG16, MONAI, OpenCV, Image Segmentation, Object Detection

Software Engineering & Full-Stack: Data Structures & Algorithms, OOP, React.js, Node.js, REST APIs, HTML5/CSS3, PHP

Databases & Cloud: MySQL, MongoDB, Vector Databases, Relational Schema Design, Google Cloud Platform (GCP)

Tools & Platforms: Git/GitHub, VS Code, Postman, Linux CLI, n8n, Google Colab, Jupyter Notebook

Security: Cybersecurity Fundamentals, Secure Coding Practices, Data Privacy Basics

PROFESSIONAL EXPERIENCE

AI/ML Engineering Intern — Unriddle Technologies (Dscribe)

Oct 2025 – Feb 2026

Shivamogga, Karnataka, India

- Designed and deployed end-to-end Retrieval-Augmented Generation (RAG) pipelines combining LLMs, vector embeddings, and semantic search over large document corpora, reducing retrieval latency and improving answer relevance.
- Applied prompt engineering strategies and fine-tuned retrieval parameters to boost Q&A accuracy, validating improvements through automated evaluation benchmarks.
- Built and maintained NLP preprocessing and text-chunking workflows for multi-document knowledge retrieval, collaborating in an Agile team environment to ship production-adjacent ML systems.
- Integrated embedding generation pipelines and semantic similarity scoring into a document intelligence product, improving output coherence across diverse query types.

Volunteer Coordinator — Anveshana Startup Summit

Sep 2024 – Present

PES Institute of Technology and Management

- Led end-to-end operations for a large-scale multi-day innovation summit, coordinating cross-functional teams and enabling high-value networking between 50+ startups and industry mentors.

HACKATHONS & ACHIEVEMENTS

1st Runner-Up — Presidency University Hackathon (48-hour sprint)

Mar 2026

- Built and presented an award-winning ML prediction dashboard, beating 100+ competing teams.

2nd Runner-Up — DIGITISE 2K25, Adichunchanagiri Institute of Technology

Oct 2025

- Architected a real-time computer vision pipeline for automated agricultural quality control, earning podium placement among 80+ teams.

Top 5 Finalist — Bapuji Institute of Technology Hackathon

May 2025

- Recognized for the Multilingual Emergency Support Platform, a real-time NLP and speech-translation system bridging language gaps between rural doctors and patients.

National-Level Participant

2025

- MONAITHON 2025 (JNNCE Shivamogga) • Agentic AI Day (Google Cloud × Hack2Skill) • GDG Solution Challenge 2025

ACADEMIC & PERSONAL PROJECTS

AI Retinal Disease Detection System	2025
- Built a multi-class retinal disease classifier using transfer learning on ResNet and VGG16, with a U-Net segmentation head for blood vessel mapping on labelled ophthalmology datasets. - Achieved 95% classification accuracy benchmarked against baseline CNNs, with full model evaluation (ROC-AUC, confusion matrices, per-class F1), demonstrating clinical-grade reliability for diagnostic support tooling.	
Multilingual Emergency Support Platform	2025
Developed a full-stack rural healthcare platform integrating real-time speech-to-text, NLP-powered multilingual translation, and LLM-based medical triage to remove language barriers between doctors and patients; Top 5 Finalist at Bapuji Institute of Technology.	
Vegetable Quality Classification System	2025
Engineered a real-time computer vision inference pipeline using image segmentation and ML classifiers to automate agricultural produce sorting across 15+ vegetable categories; 2nd Runner-Up, DIGITISE 2K25.	
Brain Tumor Segmentation (BRATS 2020)	2025
Built a 3D U-Net model to identify tumor regions from multi-modal MRI scans using the BRATS 2020 dataset.	
MIT Prediction Dashboard	2026
Built an interactive ML dashboard converting raw data into clear predictions; awarded 1st Runner-Up at the Presidency University Hackathon.	
Fitness & Calorie Tracker — Full-Stack Web App	2024
Built a production-ready full-stack application with a RESTful API and React.js dashboard for meal logging, calorie goal tracking, and fitness analytics, backed by a normalised relational schema.	

CERTIFICATIONS

- Cloud Computing Fundamentals — IBM SkillsBuild (Mar 2026) - Become a Deep Learning Engineer: Introduction to PyTorch — SkillUp by Simplilearn - AI Agents for Beginners — SkillUp by Simplilearn - MongoDB Developer Track: CRUD, Aggregation, Indexes, Data Modeling & Transactions — MongoDB University - PHP for Beginners — Great Learning Academy - Java Software Developer (Grade A) — Infosys Springboard Livelihood Program, Infosys Foundation & ICT Academy (Jul 2026)	
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EDUCATION

Bachelor of Engineering (B.E.) — Artificial Intelligence & Machine Learning	Jan 2023 - Jan 2027 (Expected)
PES Institute of Technology and Management, Shivamogga CGPA: 7.35 / 10	
Pre-University Education (PCMCs / Science)	Jan 2022 - Jan 2023
DVS Composite PU College, Shivamogga 68.16%	

LANGUAGES

English (Fluent) | Hindi (Proficient) | Kannada (Native)